🧪 **JavaScript Lab Exercises**

Topics: Operators | Conditionals | Loops | Arrays

🟢 **LEVEL 1: EASY LAB EXERCISES (Foundations)**

🟢 **LAB 1: Simple Calculator (Operators)**

📘 **Scenario**

Create a basic calculator that performs arithmetic operations.

🎯 **Requirements**

1. Declare two numbers
2. Perform:
    - Addition
    - Subtraction
    - Multiplication
    - Division
3. Display results using console.log

📌 **Concepts Used**

- Arithmetic operators
- Assignment operators

🟢 **LAB 2: Even or Odd Checker (Operators + Conditionals)**

📘 **Scenario**

Check whether a given number is even or odd.

🎯 **Requirements**

1. Declare a number
2. Use modulus operator %
3. Print **Even** or **Odd**

🟢 **LAB 3: Voting Eligibility (Conditional Statement)**

📘 **Scenario**

Check if a person is eligible to vote.

🎯 **Requirements**

1. Declare age variable
2. If age ≥ 18 → Eligible
3. Else → Not Eligible

📌 **Concepts Used**

- if...else
- Comparison operators

🟢 **LAB 4: Display Numbers 1 to 10 (Loop)**

📘 **Scenario**

Print numbers from 1 to 10.

🎯 **Requirements**

1. Use for loop
2. Display numbers in order

🟢 **LAB 5: Array Traversal (Arrays)**

📘 **Scenario**

Display all elements of an array.

🎯 **Requirements**

1. Create an array of 5 colors
2. Print each color using a loop

🟡 **LEVEL 2: MEDIUM LAB EXERCISES (Logic Building)**

🟡 **LAB 6: Student Grade Calculator**

📘 **Scenario**

Calculate grade based on marks.

🎯 **Requirements**

1. Store marks in an array
2. Calculate total and average
3. Assign grade using if...else if

📌 **Grade Logic**

- ≥ 90 → A
- ≥ 75 → B
- ≥ 60 → C
- < 60 → Fail

📌 **Concepts Used**

- Arrays
- Loops
- Conditional ladder

🟡 **LAB 7: Multiplication Table Generator**

📘 **Scenario**

Generate a multiplication table.

🎯 **Requirements**

1. Input a number
2. Print table up to 10
3. Use for loop

🟡 **LAB 8: Shopping Cart Total**

📘 **Scenario**

Calculate total shopping bill.

🎯 **Requirements**

1. Store item prices in an array
2. Calculate total using loop
3. Apply 10% discount if total > 1000

📌 **Concepts Used**

- Arrays
- Operators
- Conditional statements

🟡 **LAB 9: Find Largest & Smallest Number**

📘 **Scenario**

Analyze numeric data.

🎯 **Requirements**

1. Store numbers in array
2. Find:
    - Largest number

    - Smallest number
3. Use loop and comparison operators

🟡 **LAB 10: Login Validation**

📘 **Scenario**

Validate login credentials.

🎯 **Requirements**

1. Predefine username & password
2. Ask user to enter credentials
3. Validate using conditionals
4. Allow maximum 3 attempts

📌 **Concepts Used**

- while loop
- Logical operators
- break

🔴 **LEVEL 3: COMPLEX LAB EXERCISES (Real-World)**

🔴 **LAB 11: Employee Salary Analysis**

📘 **Scenario**

Analyze employee salary data.

🎯 **Requirements**

1. Store employee salaries in array
2. Calculate:
    - Total salary

    - Average salary
3. Display salaries above average

📌 **Concepts Used**

- Arrays
- Loops
- Conditional checks

🔴 **LAB 12: Prime Number Finder**

📘 **Scenario**

Identify prime numbers in a range.

🎯 **Requirements**

1. Define start and end range
2. Find all prime numbers
3. Count total primes

📌 **Concepts Used**

- Nested loops
- break
- Conditional logic

🔴 **LAB 13: Array Statistics Dashboard**

📘 **Scenario**

Generate analytics from numeric array.

🎯 **Requirements**

1. Store numbers in array
2. Find:

    - Sum
    - Average
    - Count of even & odd numbers
3. Display formatted report

🔴 **LAB 14: Menu-Driven Program**

📘 **Scenario**

Create a menu-driven application.

🎯 **Requirements**

1. Display menu options:
    - Add element
    - Remove element
    - Display array
    - Exit
2. Perform actions based on user choice

📌 **Concepts Used**

- do...while
- switch
- Arrays
- Operators

🔴 **LAB 15: Sales Performance Analyzer (Real World)**

📘 **Scenario**

Analyze weekly sales data.

🎯 **Requirements**

1. Store daily sales in array
2. Find:
    - Best day
    - Worst day
    - Total weekly sales
3. Display summary report